



[GCT555 3DID]

ProxiWall: Distance-Driven Semantic Zoom for HMD-Free Virtual Museum Exploration on Wall Display

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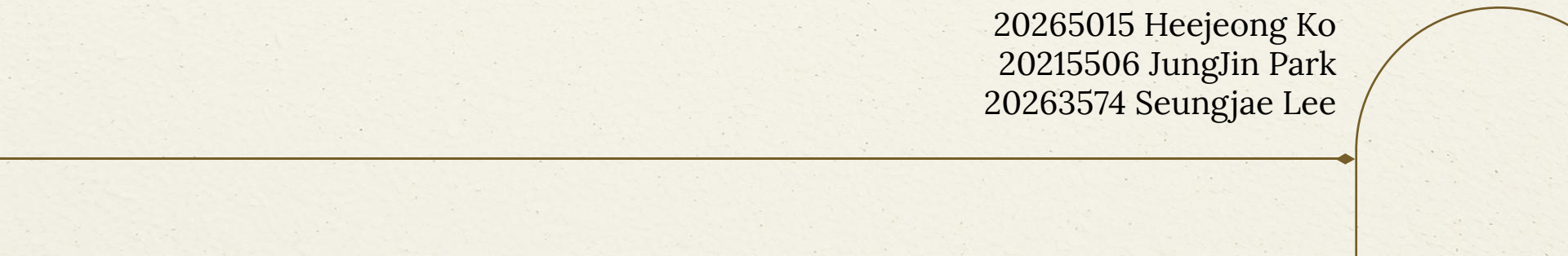


Table of contents

01

Recap

02

Study Design

03

Results

04

Deliverables

Problem & Need : UX Scenario

01 Recap

- General museum visitors need an accessible, shared, and intuitive way to explore virtual museum spaces without wearing HMDs or using controllers.
- **Problem:** A fixed wall display has only one viewpoint — yet far-viewers and near-viewers each want a different one. Without an HMD, today's wall systems cannot serve both.



FAR • Overview — visitors scan the room, taking in many works at a glance.



NEAR • Detail — visitors stop, focus, and examine a single work.

► **[The core idea]:** Museum visitors should be able to explore virtual galleries the way they explore real ones — by walking and looking — without strapping on a headset.

Collaborative Interaction

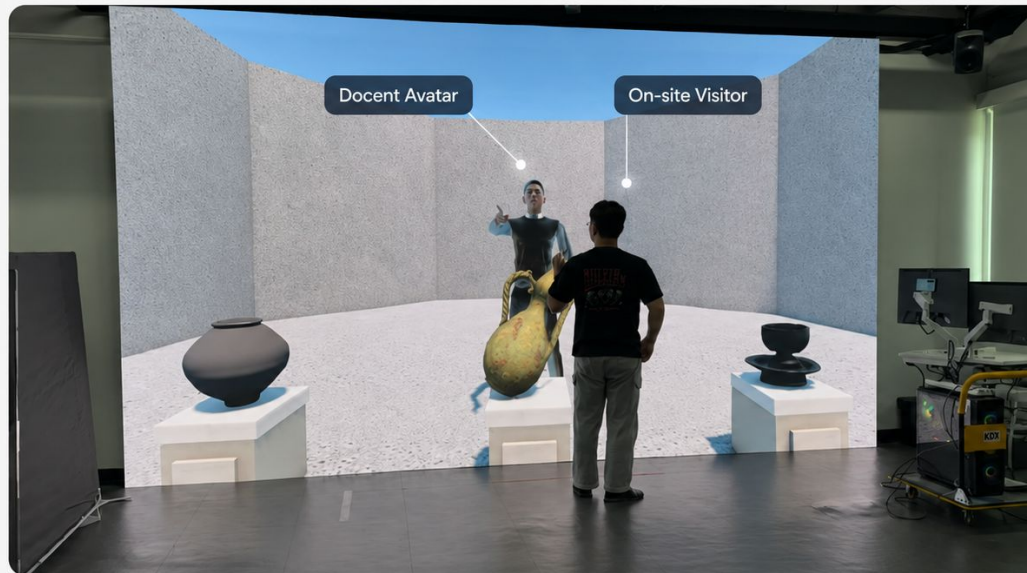
01 Recap

Docent Avatar Presence

A docent avatar appears co-present in the exhibition space, guiding the on-site visitor through both close-up artifact viewing and full-space observation.

Explore Together

The on-site visitor can explore the exhibit up close or step back to observe the overall space and the docent's movements.



Up Close

Examine the exhibit the avatar is pointing to or explaining.



Step Back

View the overall space and the docent's movements.



Approach

01 Recap

- **Environment setting**

- a. **Input** RGB Camera (Top view)
- b. **Output** 3D Rendered Video — LED Wall

- **Interaction feature**

- a. **Distance-based Semantic Zoom** : User distance from the wall automatically changes the view mode
 - i. Distance controls Overview → Browsing → Detail transition
- b. **Body-based Gallery Navigation**: Lateral movement controls horizontal exploration
- c. **Pseudo touch-based selection** : Touch confirms the final artifact selection

Rationale : Interaction Parameter

01 Recap

[Why each boundary is justified]

Cognition × Device limits × Quantified evidence

Each mode fixes **depth**, **FoV**, **visual angle** & **object count (n)** from vision-science thresholds — on a wall display, no HMD, with direct touch to commit.

Attention funnel: 7 → 3-6 → 1 object

1 Overview

~2.0 m
FoV 53°
n = 7



2 Browsing

0.7–2.0 m
FoV 53–110°
n = 3–6



3 Detail

≤0.7 m + wall touch
FoV 110°
n = 1 of 3



Wall Display

Distance (m)

① Overview ~2.0 m



Cognition peripheral + UFOV scan; n=7 within Miller 7±2.

Quantified FoV 53° · each object ~5–7° visual angle (floor-recognizable).

Greenberg Proxemics 2011 · Miller 1956 · Vogel & Balakrishnan 2004

② Browsing 0.7–2.0 m



Cognition foveal–parafoveal compare; choices cut via Hick–Hyman $\log_2(n+1)$.

Quantified FoV 53–110° · n=3–6 fits a 1–2 saccade cycle.
Miller 1956 · Hick 1952 · Hyman 1953

③ Detail ≤0.7 m + touch



Cognition 1 of 3 objects subtends >50° → foveal / ventral identification; RT=0 at n=1.

Device + motor wall-touch commit, Fitts target width maximized.

Goodale & Milner 1992 · Fitts 1954 · Dourish 2001

Comparisons

01 Recap

	ProxiWall	CollabVisAdapt (2026/VR)	BlanchTouch (2026/VR)	Co-Living between Reality & Virtuality (2026/CHI)
Main Focus	Collaborative 3D exploration	Distance-adaptive MR visualization	Vision-based wall touch	HMD-free virtual-physical integration
Interaction	Gesture + pseudo touch + sound	Distance-based adaptation	Fingertip touch detection	Natural body/touch interaction
Core Idea	Shared ↔ Private switching	Visualization switching	Surface touch sensing	Reality-virtuality coexistence
Key Difference	Multimodal collaborative workflow	Rendering adaptation only	Touch input only	Conceptual framework
Display	Dual LED wall	HMD AR/VR	HMD + RGB camera	Spatial / wall displays

Condition

02 Study Design



ProxiWall(Ours)

Naturally browse by stepping aside



Manual Touch Exploration

Touch to explore the artifacts

Research Question

02 Study Design

Main RQ:

How can **distance-based body input** and **direct wall touch** be designed and evaluated as meaningful interaction modalities for the Overview → Browsing → Selection funnel of a wall-display virtual museum?

- **Sub-RQ 1 – Interaction Usefulness / Task Performance** : Does the calibrated distance-based interaction improve artifact search and selection compared with a baseline wall-display interaction?
- **Sub-RQ 2 – Overall Experience** : How does distance-based body input affect workload, spatial presence, in a wall-display virtual museum?

Hypotheses

- H1** ProxiWall will achieve **faster task completion time** than Manual Touch Exploration.
- H2** ProxiWall will achieve **equal or higher task success rate** than Manual Touch Exploration.
- H3** ProxiWall will reduce **perceived workload** compared to Manual Touch Exploration.
- H4** ProxiWall will be perceived as more **useful and spatially engaging** than Manual Touch Exploration.

Measurement

02 Study Design

Objective Measures

Metric	Measurement
Task Completion Time	Time from Task Start to Target Artifact Selection (seconds)
Mode Usage Analysis, Mode Transition Analysis	Time spent in each interaction mode and frequency of transitions between modes during task completion
Selection Accuracy (Task Success Rate)	Percentage of trials in which the participant correctly selected the target artifact
Candidate Exploration Efficiency	Candidate changes and detail-view entries required before target confirmation

Subjective Measures

Metric	Measurement
NASA-TLX	Workload, mental demand, and physical demand
System Usability Scale(SUS)	The overall perceived usability of the system after participants complete the task.
Presence (MEC-SPQ, 8 items)	Spatial situation model, self-location, and possible actions in the virtual museum

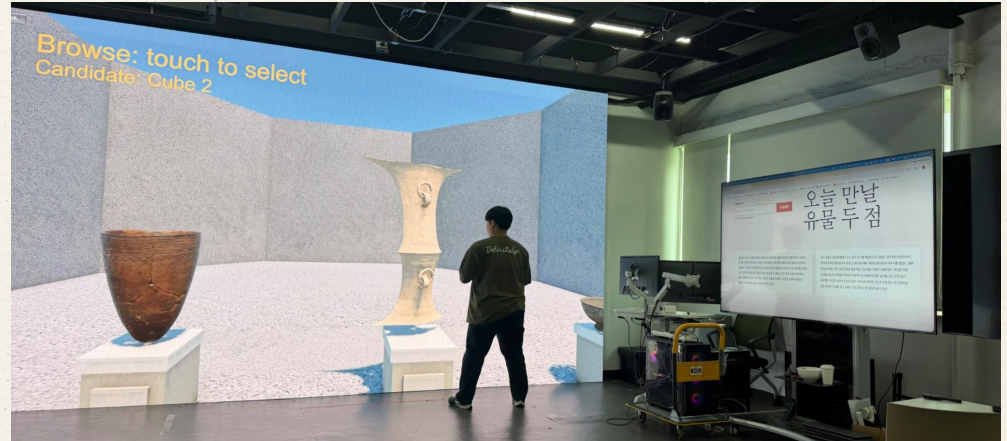
Qualitative Evaluation

Semi-Structured Interview	Naturalness, immersion, discomfort, and <u>preference</u>
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Procedure

02 Study Design

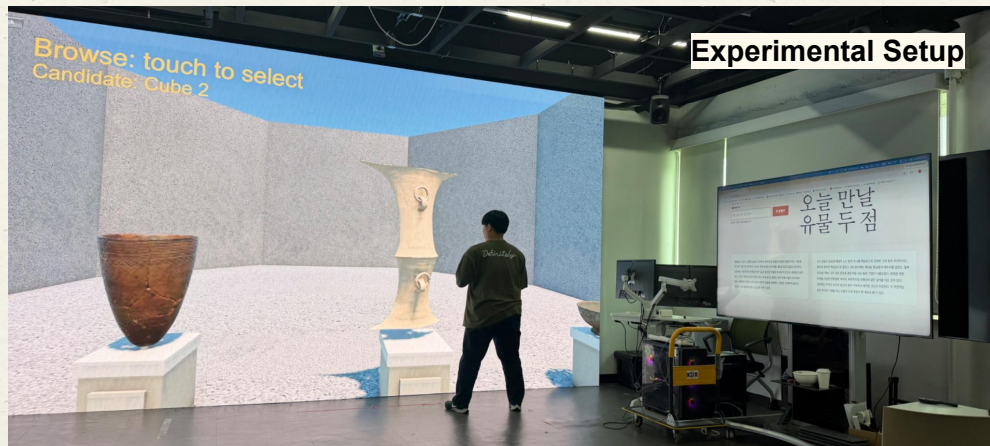
- Introduction -> Practice Session(Both Condition) -> Condition 1 X 3 task -> Questionnaire -> Condition 2 X 3 task -> Questionnaire -> Final Interview
- Task: Find the two artifacts described on the TV screen(w/o order)
- Total 18 artifacts, randomly shows 7 artifacts in each task
- Verbal confirmation by participant



Participants

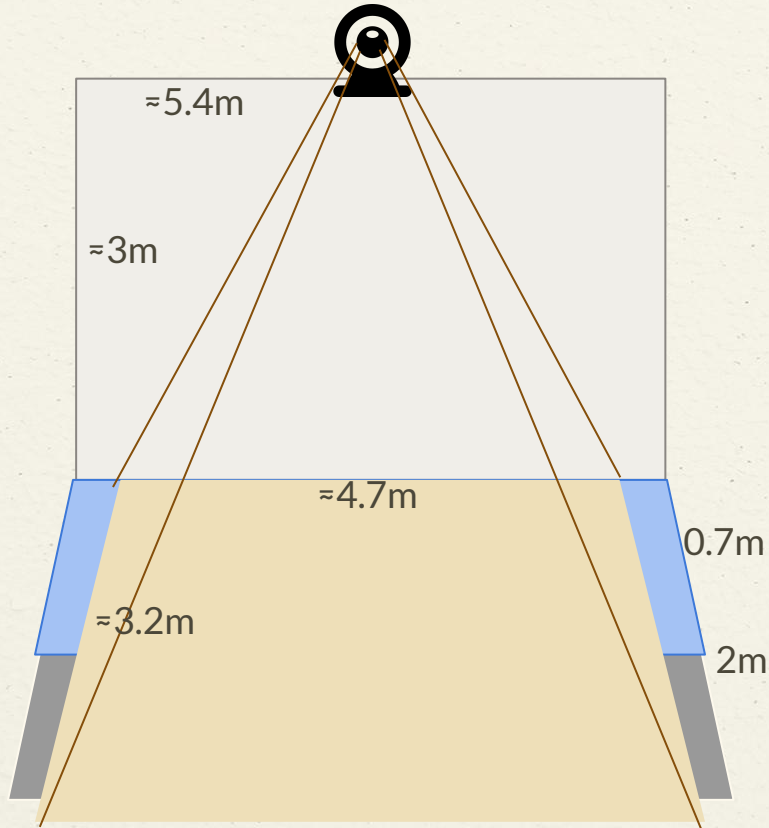
02 Study Design

- N = 6
- Age: M = 25.33, SD = 1.21
- All participants experienced both conditions:
 - ProxiWall (Ours) / Manual Touch Exploration
- After completing all conditions, participants provided:
 - Semi-structured interview feedback (Preference responses)



System Setup

02 Study Design



-  = Overview Mode
-  = Browsing Mode
- (Wall Touch → **Detail mode** for selected candidate)
-  = LED Wall
-  = Captured Area



Azure Kinect : captures ~4.7 m FOV (120°)

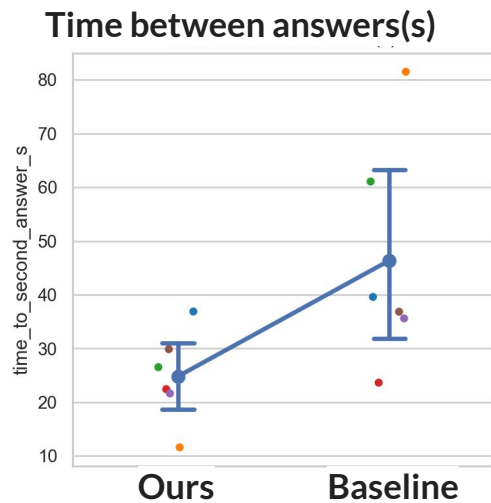
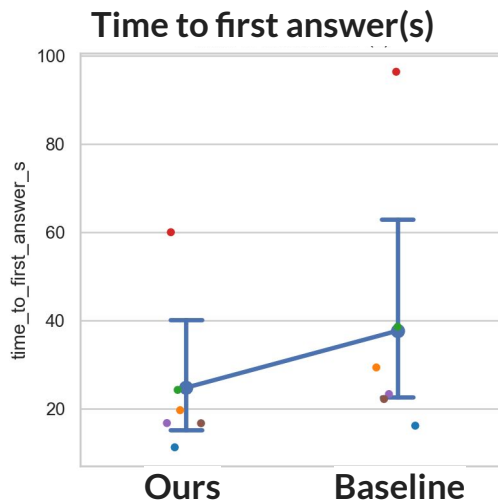
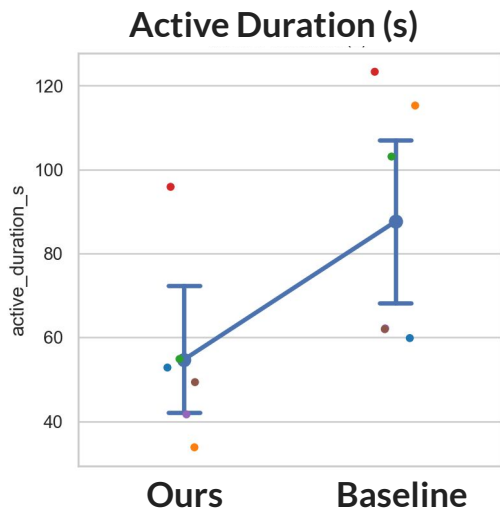


Quantitative Results

03 Results

H1. ProxiWall will achieve *faster task completion time* than Manual Touch Exploration.

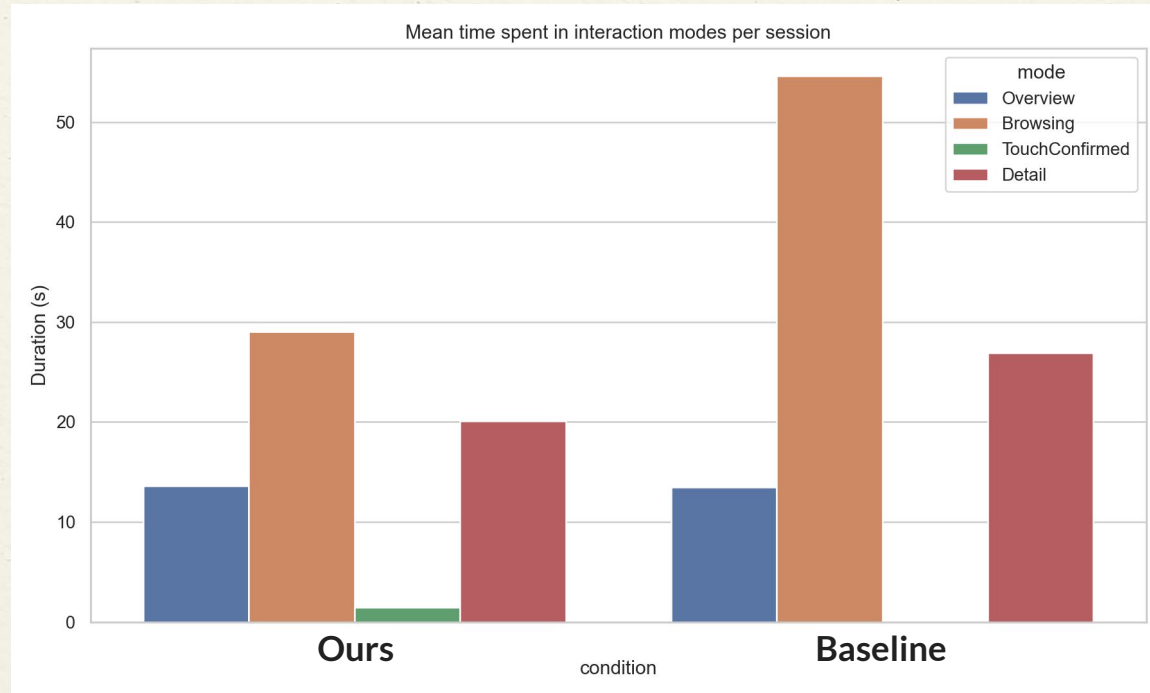
Ours **24.80s** vs Baseline 42.02s, $p=.026 \rightarrow$ **Accept**



Quantitative Results

03 Results

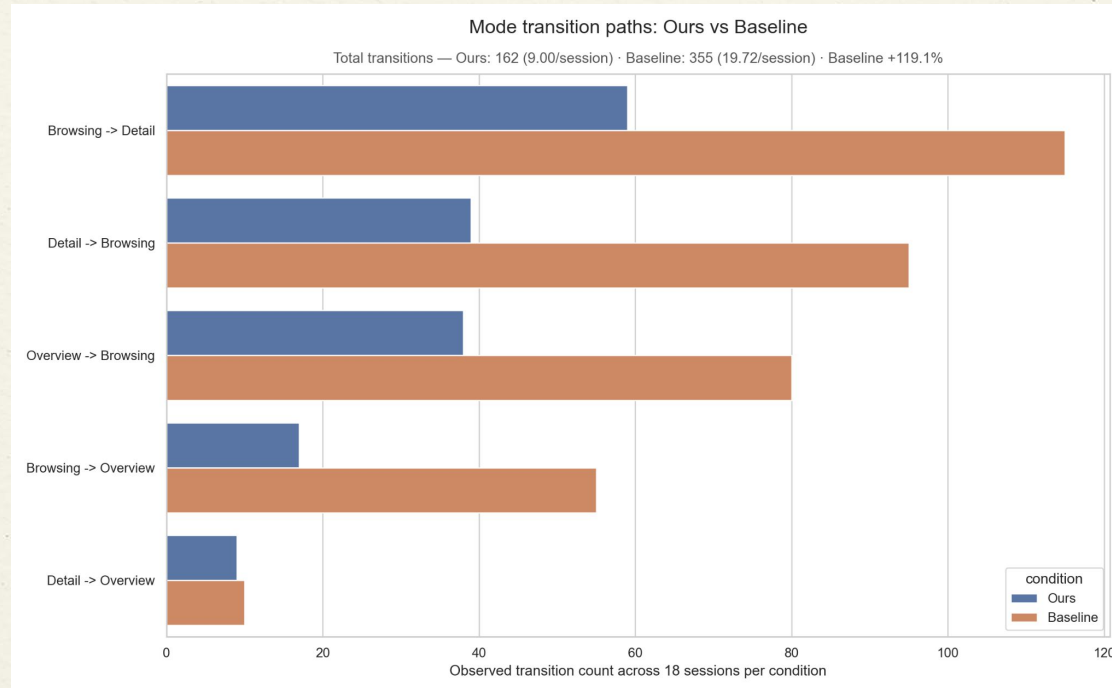
Mode Usage Analysis



Quantitative Results

03 Results

Mode Transition Analysis



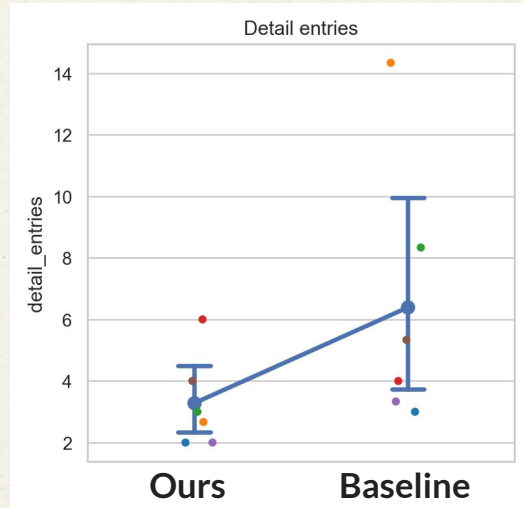
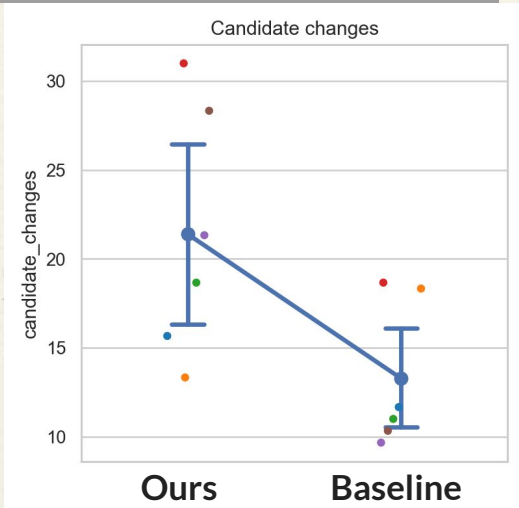
Quantitative Results

03 Results

H2. ProxiWall will achieve *equal or higher task success rate* than Manual Touch Exploration.

Both 100% , 36/36 correct → Accept with Ceiling effect

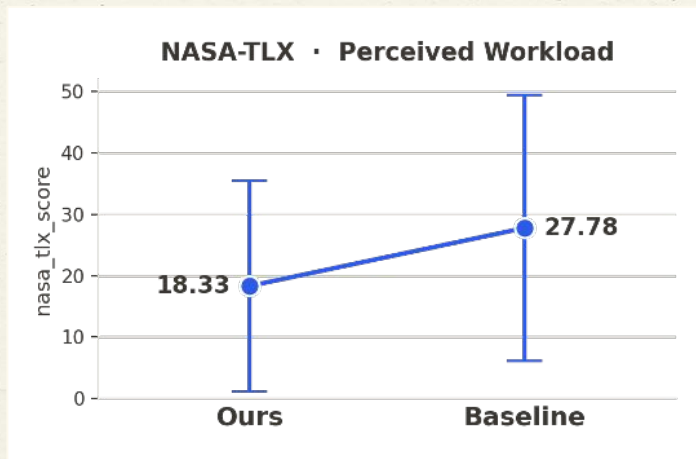
• Candidate Exploration Efficiency



Qualitative Results

03 Results

H3. ProxiWall will reduce *perceived workload* compared to Manual Touch Exploration.



Move(Ours) 18.33 vs Touch 27.78 · n.s.

→ Descriptively supported

34% lower workload than Touch

- ProxiWall 18.33 ± 17.17
- Manual Touch 27.78 ± 21.65

Lower NASA-TLX = less perceived effort.

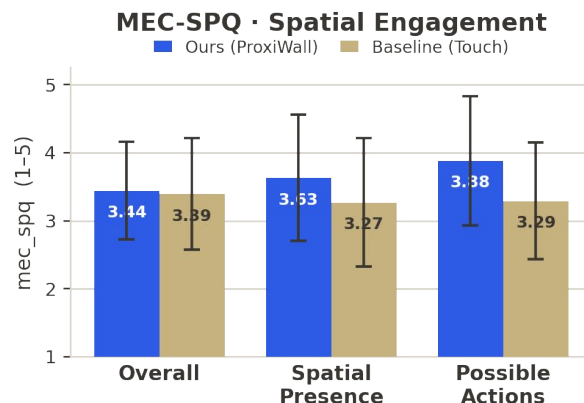
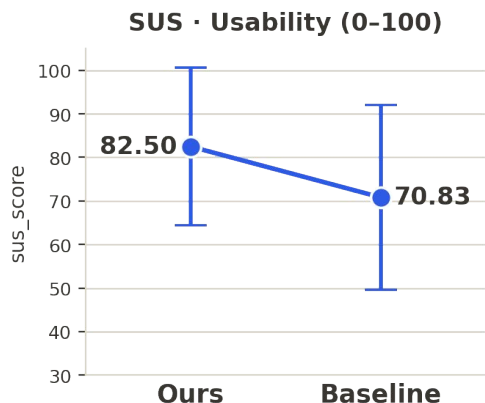
Wide, overlapping SD → descriptive trend, not statistically significant.

Qualitative Results

O3 Results

H4. ProxiWall will be perceived as more **useful[1]** and **spatially engaging[2]** than Manual Touch Exploration.

Move(Ours) SUS **82.50** vs 70.83 • spatial ↑ → **Partially supported**



[1] Bangor, A., Kortum, P. T., & Miller, J. T. (2008). An empirical evaluation of the system usability scale. *Intl. Journal of Human-Computer Interaction*, 24(6), 574-594.

[2] Vorderer, P., Wirth, W., Gouveia, F. R., Biocca, F., Saari, T., Jäncke, L., ... & Klimmt, C. (2004). *Mec spatial presence questionnaire*. Retrieved Sept, 18(2004), 2015.

Semi-Structured Interview

03 Results

5/6 participants preferred Ours (move/walk)

Ours · move/walk- body navigation

■ Worked well

- Immersion & presence — direct interaction with the media wall; *“felt present inside the virtual museum.”* P3, P6
- Natural & fun — enjoyed walking to view; motion navigation more fun & engaging. P4, P5
- Easy to use — *“far more convenient and easier than touch.”* P2

■ Friction

- Movement-view proportion unclear; left/right speed hard to control. P1
- Slight dizziness from repeated size/zoom changes. P2
- Large, moving artifacts made aiming hard. P5

Baseline · touch-direct selection

■ Worked well

- Less tiring than moving around oneself. P1
- Convenient for pulling up information by hand. P6

■ Friction

- Repetitive — *“had to touch every time,” “too much back-and-forth.”* P2, P3
- Hard to separate selection from movement. P4
- Touch errors / misfires were uncomfortable. P6
- Unclear necessity — *“walking > touch.”* P5

Takeaway

Move / Walk was preferred for immersive and spatially natural exploration, while Touch was mainly useful as a fallback for precise confirmation.

Discussion

04 Deliverables

- **Efficiency:** ProxiWall reduced task completion time compared to Manual Touch Exploration.
- **Lower burden:** NASA-TLX showed a lower workload tendency for ProxiWall.
- **Usability & engagement:** SUS and MEC-SPQ results suggest higher perceived usability and stronger spatial-action engagement.

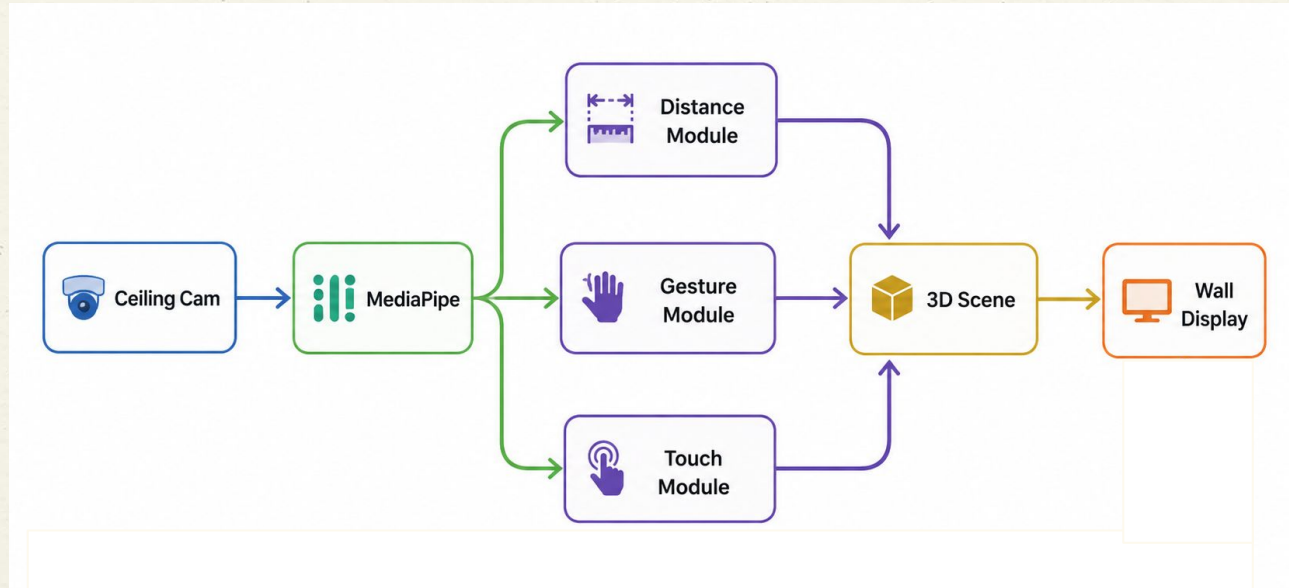


- **Socially meaningful:** HMD-free wall interaction supports accessible and shared museum exploration.
- **Design implication:** Lead with body-based navigation; keep touch for explicit information access, final confirmation, and error recovery.



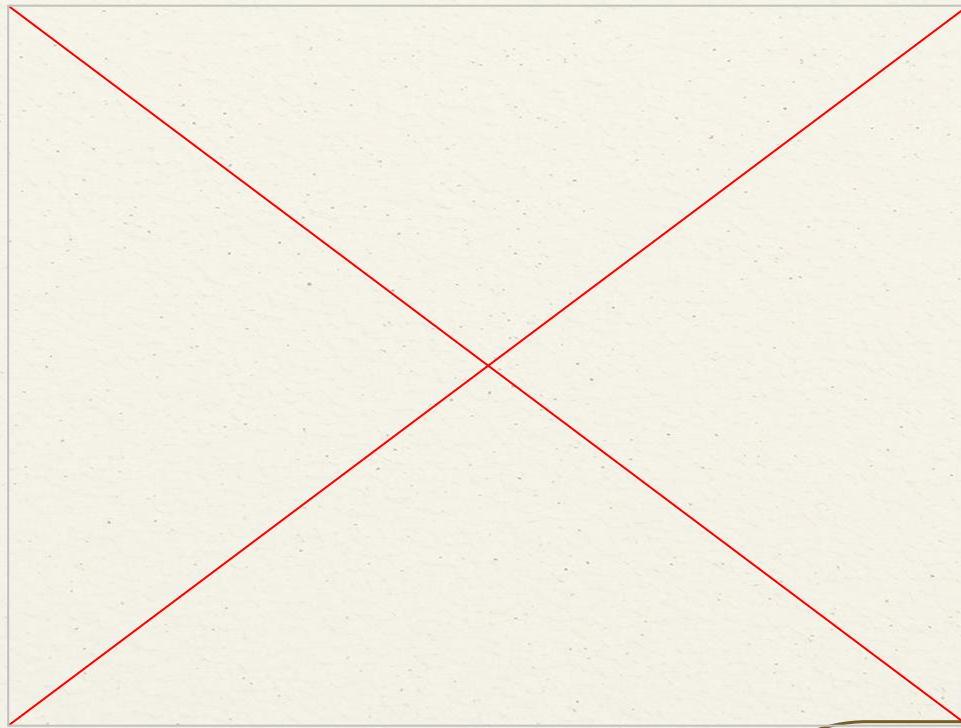
Interaction Flow

04 Deliverables



Demo Video

04 Deliverables



Thanks!

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